Adaptive façade-tilted bird's-eye view

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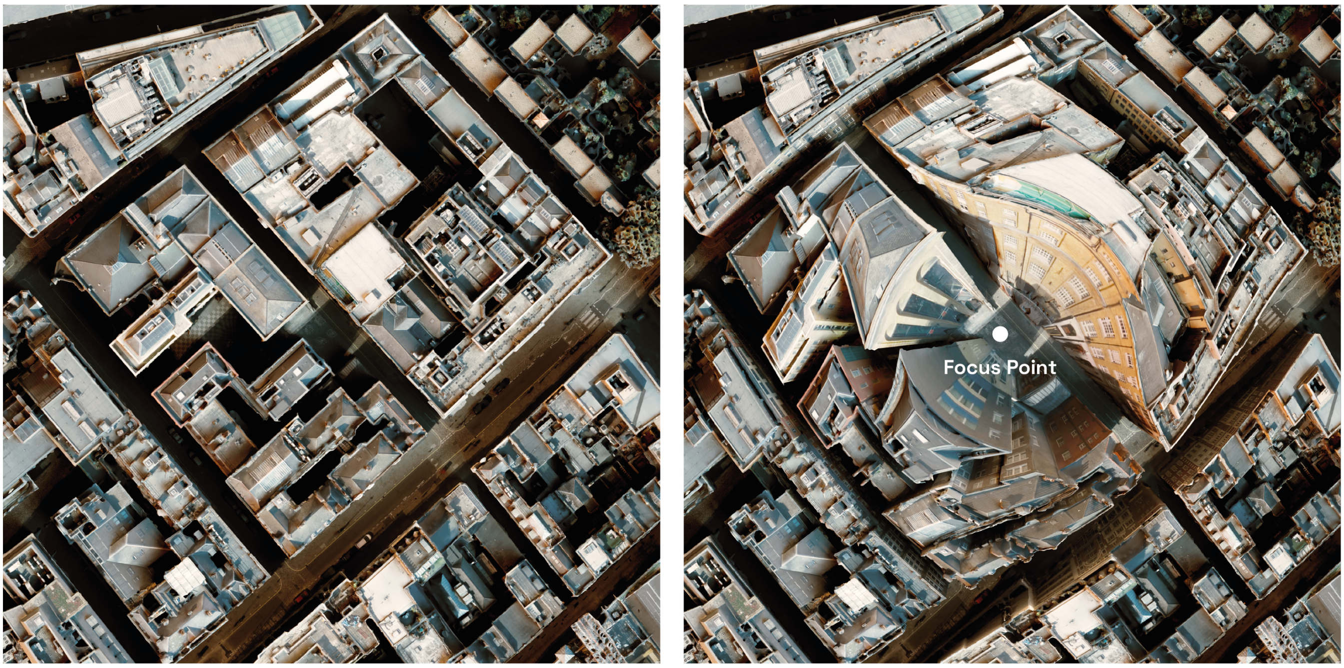

Figure 1: *Traditional bird eye view (left) vs. our adaptive façade-tilted view with focus point (right)*

Abstract
Urban pedestrian navigation faces a trade-off between overview and detail. Traditional 2D maps provide spatial overview but conceal façade features essential for wayfinding. 3D perspective maps enhance self-localization yet sacrifice spatial context due to occlusion. Current solutions force users to switch between multiple representations. We introduce adaptive façade tilting, a focus+context technique applying height-dependent radial displacement to 3D urban models, dynamically tilting building façades outward from a user-defined focus point. This reveals occluded vertical surfaces while preserving ground-level references for spatial orientation. Our evaluation with 27 participants shows that façade-tilted representations better align with natural navigation strategies (17.8% vs. 42.4% deviation) and generate higher user confidence despite comparable accuracy. Our findings suggest adaptive façade tilting bridges cognitive spatial processing and digital map representation.

1. Introduction

A pedestrian emerges from a subway station in an unfamiliar city, smartphone displaying a navigation app. Despite GPS coordinates and detailed maps, they cannot determine which direction they face or align the abstract screen representation with their concrete environment.

Urban pedestrian navigation suffers from a trade-off between overview and detail. Traditional 2D bird's-eye view maps provide efficient spatial overview but conceal critical façade features essential for wayfinding [TD14]. Conversely, 3D perspective maps enhance self-localization but sacrifice spatial context and suffer from occlusion [HLC12,LTDJ08]. Clustered buildings often occlude features, forcing repeated interactions rather than focusing on orientation [DZMQ15].

This suppresses natural navigation strategies, as people instinctively orient using architectural landmarks and façade features [KYLK23, DZMQ15], yet conventional bird's-eye views systematically conceal these navigation-critical cues. Existing multi-perspective approaches introduce distortions that complicate real-world alignment [TD14, PTD13], while interactive methods target only isolated elements [DZMQ15].

We introduce **adaptive-tilted bird's-eye view**, a novel focus+context technique using height-dependent radial displacement. Our approach applies radial displacement outward from a user-defined focus point (typically the user's current location) to 3D urban models, revealing landmark features such as façade details while preserving ground-level street layout. The technique locally reveals vertical surfaces with 360-degree visibility without global distortion, while maximizing screen space efficiency by minimizing the footprint of navigationally irrelevant rooftops (Figure 5). This maintains the spatial overview benefits of bird's-eye view while ensuring compatibility with existing smartphone platforms through its 2D representation of 3D spatial data.

Evaluation with 27 participants demonstrates better alignment with natural navigation behavior and higher user confidence despite comparable accuracy rates.

1.1. Contributions

(1) Novel adaptive façade tilting technique providing comprehensive architectural visibility while preserving spatial context, and (2) empirical validation demonstrating improved cognitive alignment with natural navigation strategies, higher user confidence, and enhanced façade feature utilization.

2. Related Work

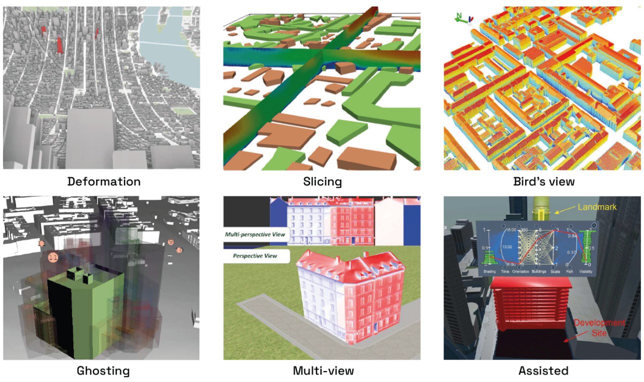

Figure 2: *Examples of approaches to handle occlusion: Deformation [CMF*22], ghosting [OSS*17], slicing [KPMB21], bird's view [WSK18], multi-view [PTD13], assisted-steering [ZZL21]. Adapted from [MOM*24].*

Researchers address occlusion handling in 3D urban visualization through diverse approaches with distinct limitations. A comprehensive survey by Miranda et al. [MOM*24] categorizes these techniques into six primary strategies: deformation, ghosting, slicing, multi-view approaches, assisted steering, and bird's view perspectives.

Deformation techniques alter 3D urban geometry through scaling and geometric transformations to reveal occluded content [DZMQ15]. These approaches represent the closest conceptual foundation to our focus+context method. However, existing deformation methods lack the comprehensive 360-degree visibility required for pedestrian navigation tasks, where users need to assess all directions for orientation decisions such as turning or reversing direction.

Multi-view approaches combine multiple perspectives [PTD13], though they necessitate perspective switching that interrupts navigation workflows and increases cognitive load during wayfinding tasks [CLW*23].

Bird's view approaches limit access to façade features that serve as important wayfinding landmarks, as research shows that eye-level signage and landmark imagery are particularly useful for pedestrian orientation and navigation [VHA*16]. Despite these limitations, bird's view remains the most popular approach in surveyed works across all use cases [MOM*24].

Several techniques target specific application domains but are not suitable for pedestrian navigation. Ghosting approaches utilize transparency for disaster management scenarios, slicing methods employ cross-sectional views for wind analysis, and assisted steering automatically computes viewpoints for predefined positions [CKS*15, KPMB21, ZZL21]. These methods either lack spatial context preservation or require specialized interaction paradigms incompatible with continuous navigation workflows. While augmented reality approaches for pedestrian navigation are explored in AR head-mounted displays [TP20], smartphone-based solutions remain the most practical platform for everyday navigation, offering ubiquitous accessibility, consistent performance across diverse hardware capabilities, minimal battery consumption, and reliable positioning even under challenging signal conditions [JCC*23].

Miranda et al. highlight that advanced occlusion-handling strategies remain underexplored [MOM*24], while systematic reviews show 3D visualization is prone to occlusion and clutter [DRST14]. Our technique addresses this gap by providing localized façade visibility without the usability challenges of full 3D representations, making it suitable for mobile navigation applications.

3. Implementation

Our adaptive façade tilting technique applies height-dependent radial displacement outward from a user-defined focus point to 3D urban models, revealing landmark features such as façade details. (Figure 3).

Figure 3: *Adaptive façade-tilted view with focus point and 62.5m influence zone.*

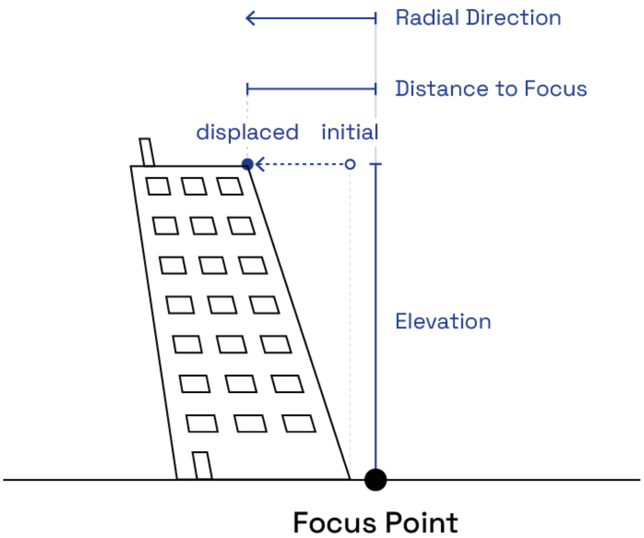

Figure 4: *The three key components of point displacement: radial direction, distance to focus, and elevation. We displace each point in the spatial influence zone of the 3D urban model based on these key components.*

3.1. Core Algorithm

Each point p in the 3D urban model is displaced according to:

$$\mathbf{D}(p) = h(p) \cdot w(p) \cdot \hat{\mathbf{r}}(p) \quad (1)$$

with quadratic distance falloff that diminishes the displacement effect with increasing distance from the focus point:

$$w(p) = \max\left(0, 1 - \left(\frac{d(p,f)}{R}\right)^2\right) \quad (2)$$

where $h(p)$ denotes elevation above ground level, $d(p,f)$ represents horizontal distance to focus point f, and $R = 62.5$ defines the spatial influence zone (Figure 4).

3.2. Design Principles

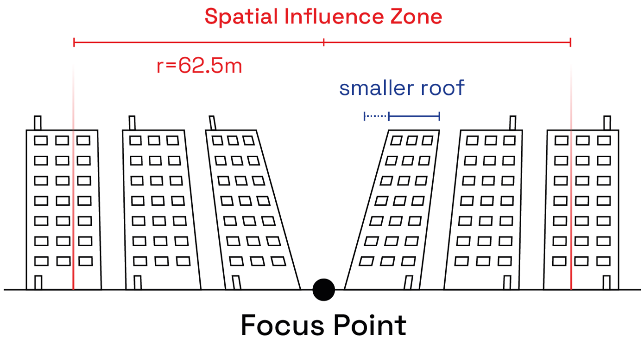

Figure 5: *Side view showing displacement of the 3D urban model within the 62.5-meter influence zone. Buildings closer to the focus point experience greater displacement.*

Displacement increases linearly with elevation, ensuring ground-level elements remain in original positions to preserve navigation references. The distance-based attenuation creates smooth transitions that prevent abrupt visual discontinuities. Since displacement is based on elevation and decreases with distance from the focus point, building rooftops are reduced in area and occupy less screen space. This is advantageous since rooftops provide limited orientation value compared to façade features (Section 5). The 62.5 influence radius corresponds to empirical studies showing landmarks achieve greatest façade visibility at this distance [BMK08]. Beyond this radius, points remain unaffected, ensuring localized focus while preserving urban context (Figure 5).

4. Study Design and Methodology

4.1. Hypotheses

We formulate four hypotheses for the tilted bird's-eye view representation:

H1 - faster task completion

H2 - higher user confidence

H3 - better alignment with natural navigation strategies

H4 - higher user preference

4.2. Study Design

We conduct a between-subjects online experiment comparing tilted and traditional bird's-eye view representations. 27 participants are randomly assigned to conditions (17 tilted bird's-eye view, 10 traditional bird's-eye view). No exclusion criteria are applied to maximize accessibility and participation rates, though demographic data are collected but not analyzed due to the limited sample size.

4.3. Materials and Task

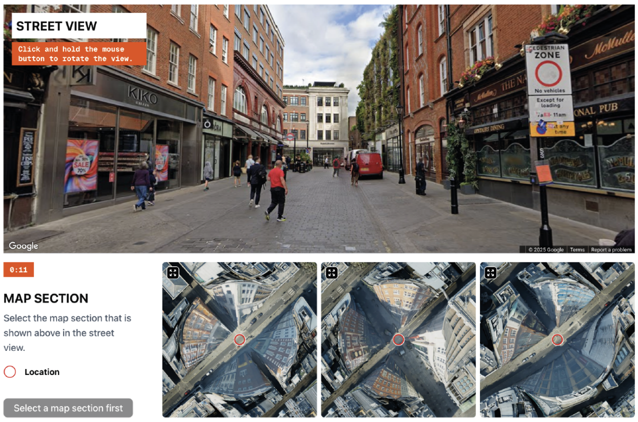

Figure 6: *Task interface: street view (top) and three map options (bottom).*

The study uses a Next.js web application (Figure 6). Google Street View API provides street-level imagery with mouse rotation as the only permitted interaction. Map sections are generated using high-resolution 3D city models rendered in Cinema 4D, displaying 125 × 125 areas.

We select five urban locations (three London, two Cambridge) with multi-story buildings and complex spatial layouts where at least three pathways diverge from each location, creating decision points that require active navigation choices. For each correct location, two distractor options are generated from the surrounding area, each similarly featuring multiple pathway intersections to ensure comparable spatial complexity across all map options.

4.4. Procedure

The study consists of three phases: demographics and navigation strategy questionnaire, map selection tasks with confidence ratings, and post-study assessment with semi-structured interviews (6 participants). Response times are tracked automatically.

5. Results

5.1. Accuracy

Both representations achieve high accuracy rates (Figure 7). Traditional bird's-eye view: 93.9%, tilted bird's-eye view: 89.4%. Users with traditional bird's-eye view complete tasks in 36.9 seconds average, while users with tilted bird's-eye view require 50.7 seconds—learning effects for the novel interface may explain this [PHH19].

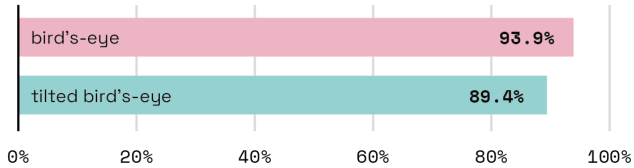

Figure 7: *Task accuracy rates by condition: Traditional bird's-eye view (93.9%) and tilted bird's-eye view (89.4%).*

5.2. User Confidence

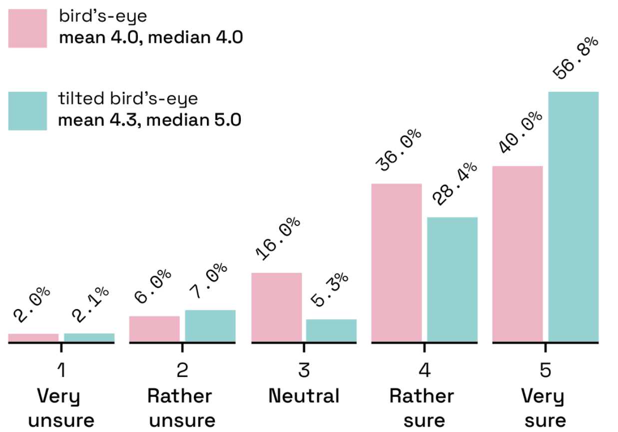

Figure 8: *User confidence ratings by condition showing distribution across 5-point Likert scale and summary statistics.*

Confidence ratings favor the tilted bird's-eye view representation (Figure 8). Traditional bird's-eye view: mean 4.06, tilted bird's-eye view: mean 4.31. Notably, 56.8% of tilted bird's-eye view responses rate maximum confidence versus 40% in traditional bird's-eye view condition.

5.3. Navigation Strategy Alignment

The tilted bird's-eye view representation demonstrates significantly better alignment with natural navigation behavior—17.8% deviation versus 42.4% for traditional bird's-eye view condition (Figure 9). In real-world navigation, participants rely on landmarks (81.5%), street layout (70.4%), and façade features (66.7%). The tilted bird's-eye view condition shows exceptional façade feature usage (94.1%), while traditional bird's-eye view condition shows minimal usage (10.0%).

5.4. Hypothesis Evaluation

H1 (Response Time): Not Supported – Tilted bird's-eye view condition requires longer completion times, potentially due to learning effects.

H2 (Confidence): Supported – Higher confidence ratings and greater proportion of maximum confidence responses.

H3 (Navigation Strategy): Strongly Supported – Dramatically

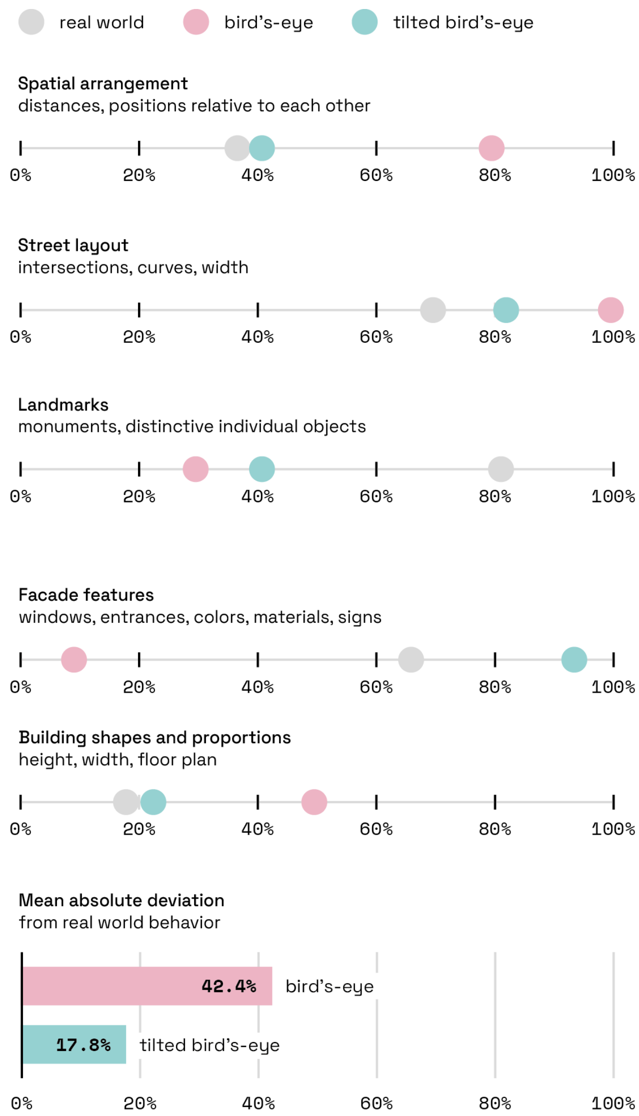

Figure 9: *Navigation strategy comparison showing feature usage patterns for real-world navigation, traditional bird's-eye view (42.4% deviation), and tilted bird's-eye view (17.8% deviation) relative to natural navigation behavior.*

better alignment with natural strategies, particularly for façade features.

H4 (User Preference): Partially Supported – Despite the unfamiliarity requiring longer task completion, users report higher confidence in their responses with the tilted bird's-eye view representation.

The results reveal that while traditional bird's-eye view representations maintain comparable accuracy and advantages in speed, the tilted bird's-eye view representation generates higher confidence and better cognitive alignment with natural navigation strategies.

6. Discussion

Our evaluation reveals that while adaptive façade tilting requires longer task completion times, it achieves superior user confidence and notably improved alignment with natural navigation strategies. This performance-preference pattern reflects the dual cognitive demands users face when simultaneously learning novel interface conventions while performing navigation tasks.

Divided attention effects can explain the extended completion time, consistent with research demonstrating that navigational aids impair spatial processing by dividing cognitive resources [GBMT13]. Users of the tilted representation process both familiar navigation cues and unfamiliar visual transformations, whereas traditional bird's-eye view users can focus entirely on established spatial reasoning patterns. Single-session usability assessments systematically favor familiar interfaces, as initial advantages diminish with extended exposure [SZUS12]. When façade details become visible, users immediately incorporate them into their navigation strategies, indicating that existing cognitive frameworks are simply inaccessible with conventional representations.

Our strategy alignment methodology—comparing interface behaviors with natural navigation patterns—demonstrates that tilted representations better match real-world navigation strategies. Users show 94.1% façade feature utilization compared to 10.0% in traditional views, more closely approximating the 66.7% façade reliance observed in natural wayfinding. This alignment suggests the interface supports rather than constrains intuitive spatial reasoning.

Adaptive façade tilting eliminates the interface-switching problem characteristic of existing multi-view approaches [PTD13], while simultaneously addressing occlusion challenges and preserving spatial context. This unified solution proves particularly valuable for dense urban environments where architectural landmarks serve as primary navigation references.

Our study design presents constraints including between-subjects sampling (n=27) and online testing limitations. The technique's effectiveness may diminish in environments with limited architectural diversity, such as rural areas or uniform building contexts. Additionally, the approach requires high-resolution 3D urban models, potentially restricting deployment to well-mapped metropolitan regions.

Future research should examine performance trajectories through longitudinal field studies with real-world navigation tasks [RHLB14]. Such studies remain essential for validating laboratory findings in authentic navigation contexts where cognitive alignment advantages may translate into practical navigation improvements over extended use periods, particularly across diverse cultural contexts where spatial navigation strategies may vary.

Adaptive façade tilting demonstrates that supporting natural spatial cognition can yield measurable benefits in user confidence and strategy alignment, even when speed metrics show initial disadvantages.

7. Conclusion

With high-resolution 3D urban models becoming increasingly available, digital navigation interfaces have new opportunities to better align with how humans naturally navigate urban environments. Our adaptive façade tilting technique demonstrates that by revealing architectural features while preserving spatial context, digital maps can better support intuitive wayfinding strategies.

The cognitive benefits—higher user confidence and dramatically improved alignment with natural navigation behavior—suggest that supporting rather than constraining human spatial cognition yields measurable advantages in interface design. While users require additional time to process the novel representation, their immediate adoption of façade features for navigation tasks indicates that existing cognitive frameworks are simply inaccessible with conventional bird's-eye views.

This work points toward a broader design philosophy: urban navigation interfaces should complement human spatial intelligence rather than substitute it. As urban environments become denser and more crowded [WWW16], techniques that bridge the gap between digital representation and embodied spatial experience will prove essential for creating navigation tools that feel natural rather than artificial. By maintaining compatibility with existing smartphone platforms through its 2D representation, our approach offers a practical solution that users can immediately adopt without specialized hardware or applications.

8. Acknowledgments

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